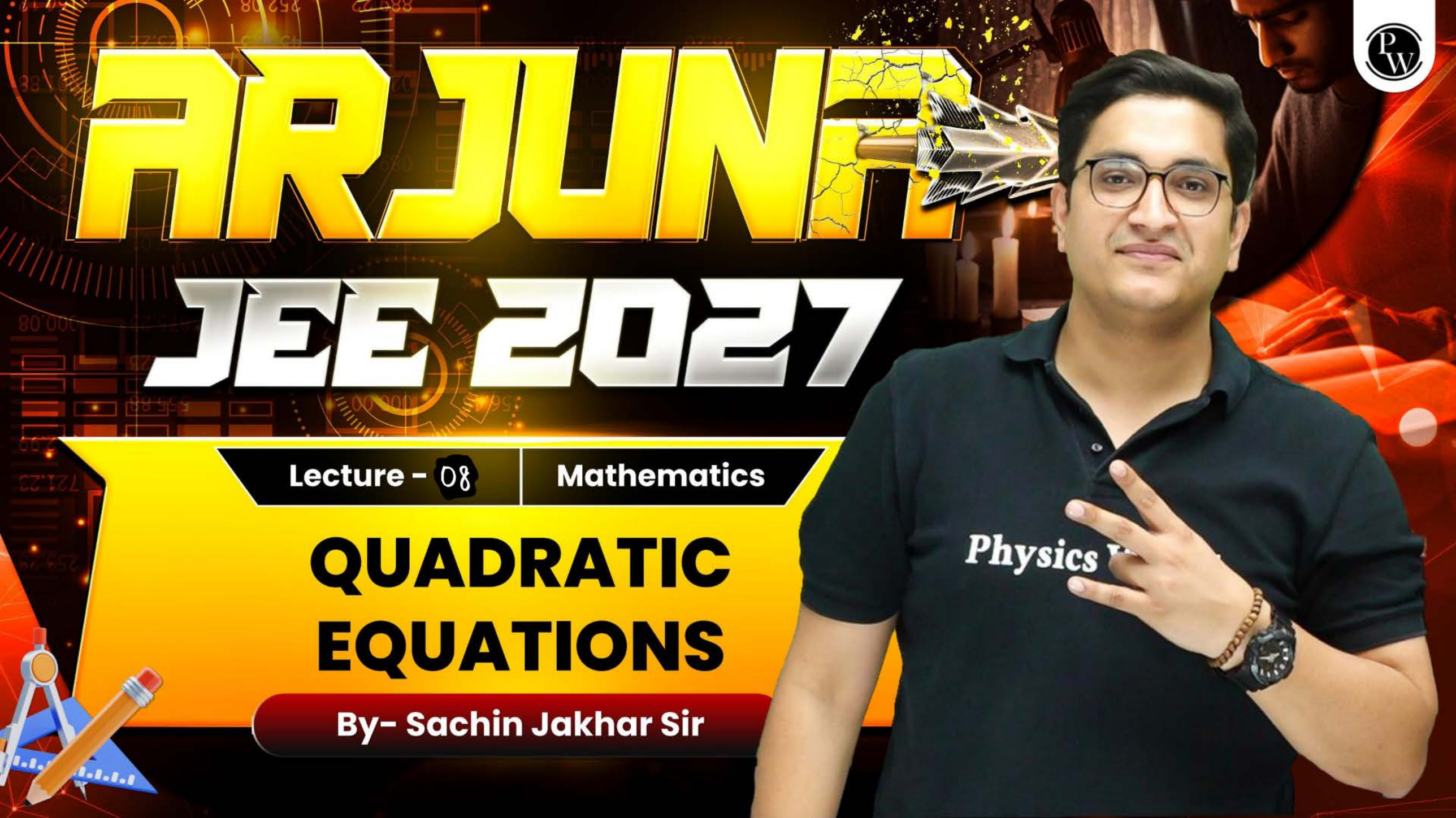
ARJUNA JEE 2027

Lecture - 08 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A Graph of Quadratic

B Range of Quadratic

C Question Practice

D

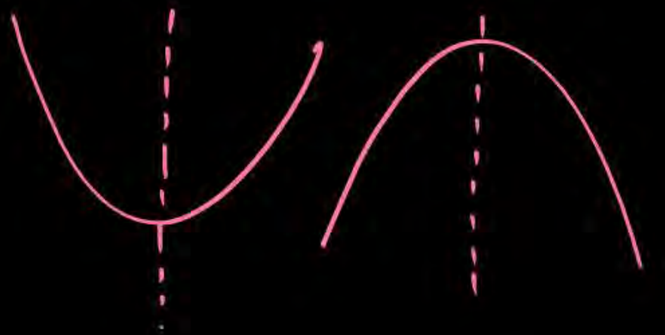


LAST CLASS REVISION

Answer the following Questions:

$$y = ax^2 + bx + c$$





(i) If $a > 0$ then graph is opening upward.

(ii) If $a < 0$ then graph is opening downward.

(iii) Coordinates of vertex is $V\equiv\left(\frac{-b}{2a}, \frac{-D}{4a}\right)$.

(iv) Point where graph cuts y-axis $(0,c)$.

(v) Point where graph cuts x-axis are Roots of quadratic.

(vi) Graph of quadratic is symmetric about $\left(x = \frac{-b}{2a}\right)$.

Question Practice

Question

If $x = \sqrt{11} + 4$ then value of $x^5 - 8x^4 + 12x^3 - 8x^2 + 11x + 5$ is _____.

$x - 4 = \sqrt{11}$

SBS

$x^2 - 8x + 16 = 11$

$x^2 - 8x + 5 = 0$

[**Remainder** $= 360x - 235$

Ans. : $360\sqrt{11} + 1205$]

Critical Thinking Question (CTQ)



Re-attempt

If both the roots of the quadratic equation $x^2 - (2n + 18)x - n - 11 = 0, n \in I$, are rational, then find the value(s) of n.

$$D = \left(-(2n+18)\right)^2 - 4 \times 1 \times (-n-11)$$

$$= \left(-2(n+9)\right)^2 + 4(n+11)$$

$$= 4(n+9)^2 + 4(n+11)$$

$$\Rightarrow D = 4\{(n+9)^2 + (n+11)\}$$

+ve/0 Perfect square

Say

$\# \quad n+9 = I_1$

$n+11 = I_1 + 2$

$$D = 4\left(I_1^2 + I_1 + 2\right) = 0, 1, 4, 9, 16 \ldots$$

P.S. P.S.

$I_1 = 1 \Rightarrow n+9 = 1 \Rightarrow n = -8$

$I_1 = -2 \Rightarrow n+9 = -2 \Rightarrow n = -11$

fnf

$I_1^2 + I + 2$

$I_1 = 0 \Rightarrow 2$ X

$I_1 = -1 \Rightarrow 1 - 1 + 2 = 2$ X

$I_1 = -2 \Rightarrow 4 - 2 + 2 = 4$ ✓

$I_1 = 1 \Rightarrow 1 + 1 + 2 = 4$ ✓

$I_1 = 3 \Rightarrow 9 + 3 + 2 = 14$ X

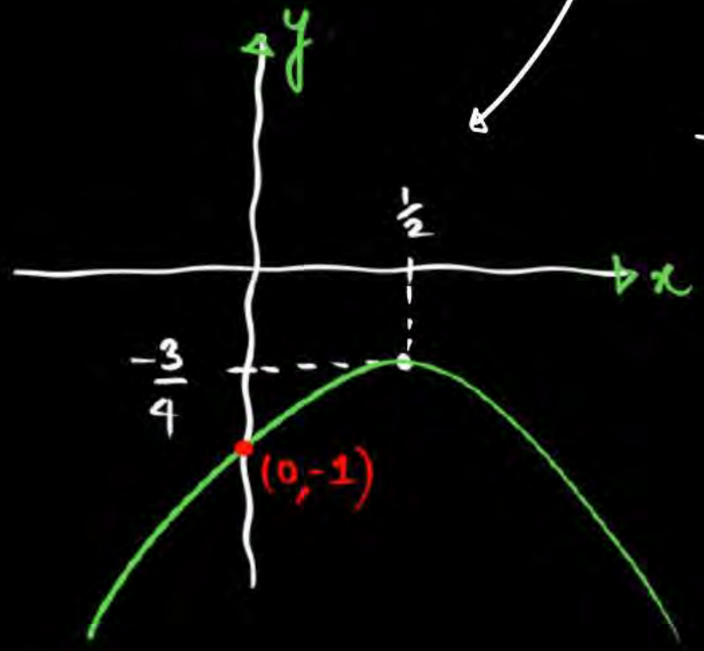
Question

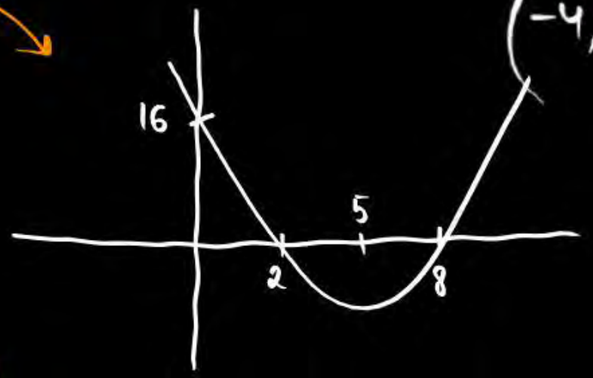
Draw graph of following quadratic :

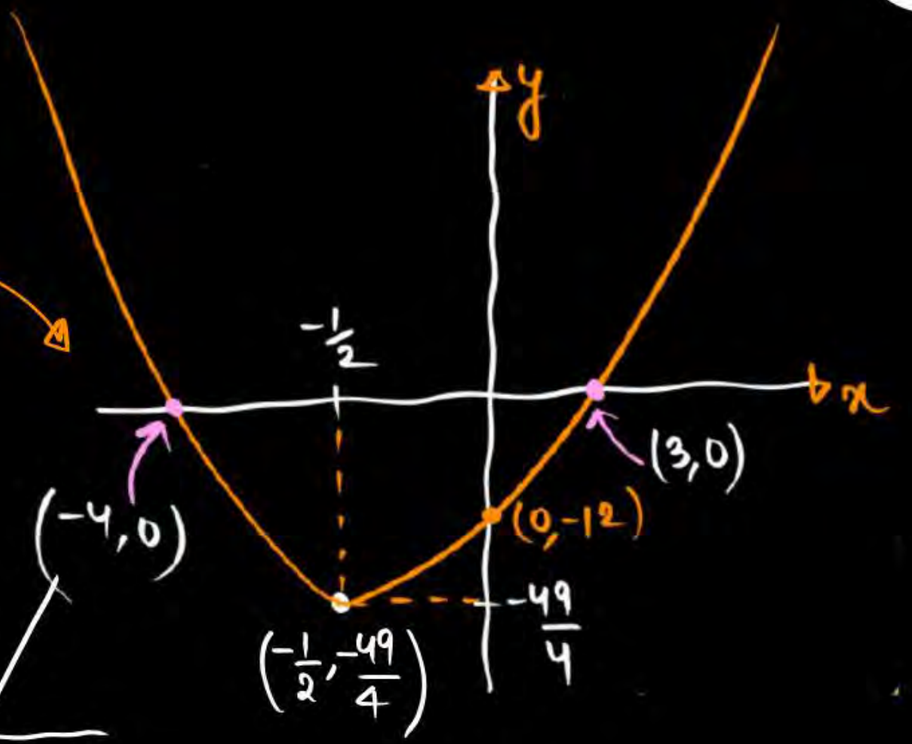
(i) $f(x) = x^2 + x - 12$

(ii) $f(x) = x - (1 + x^2) = \boxed{-x^2 + x - 1}$

(iii) $f(x) = x^2 - 10x + 16$







Acidic Question

The graph of $y = ax^2 + bx + c$ is shown in the figure, then which of the following is(are) correct?

A $ab^2c^3 < 0$

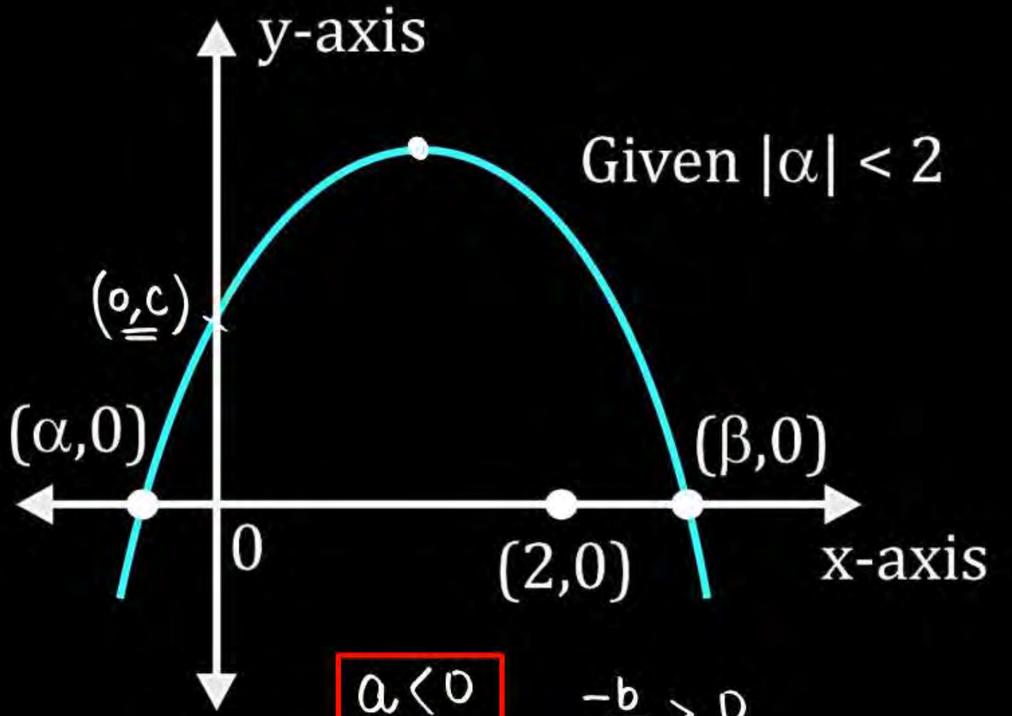
B $ab < 0$

C $bc(4a + 2b + c) > 0$

$x = 2$

$y = 4a + 2b + c$

(value at $x = 2$) ***



$a < 0$

$c > 0$

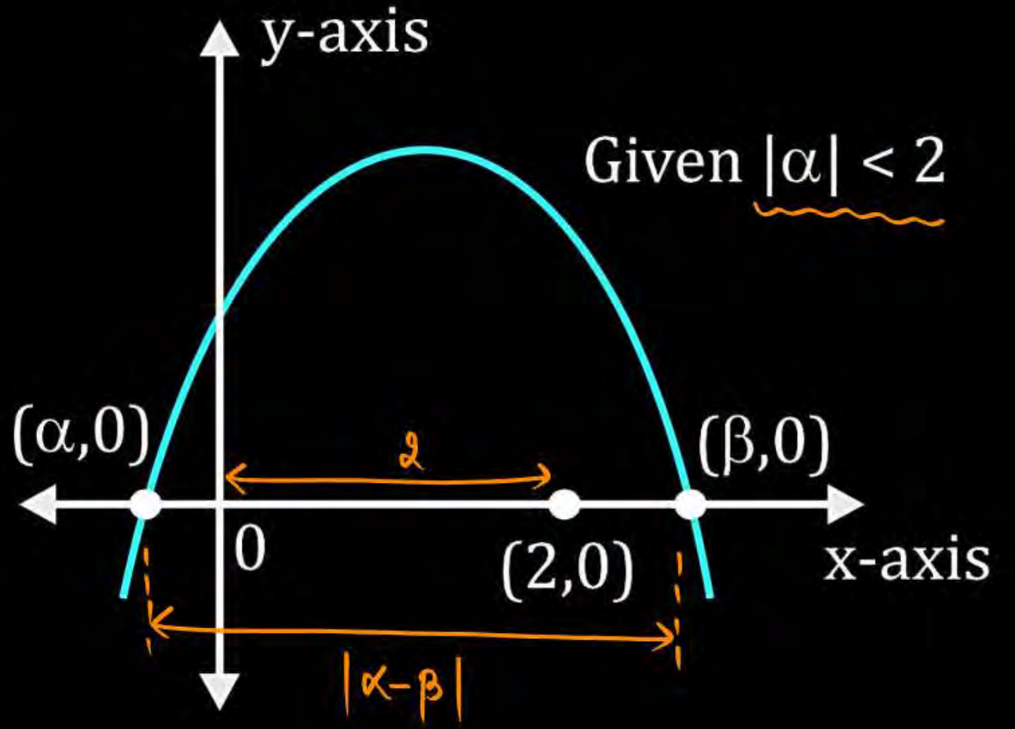
$\frac{-b}{2a} > 0$

$-b < 0$

$b > 0$

Acidic Question

The graph of $y = ax^2 + bx + c$ is shown in the figure, then which of the following is(are) correct?



D $ab(4a - 2b + c) > 0$

- a: $\ominus$ $\oplus$
- b: $\ominus$ $\oplus$
- value at $x=-2$: $\ominus$
- $x=-2$
- $\# y = 4a-2b+c$

E $b^2 > 4(a^2 + ac)$

$b^2 > 4a^2 + 4ac$

$b^2 - 4ac > 4a^2$

$\# \frac{b^2-4ac}{a^2} > 4$ ✓

F $ax^2 + bx + c > 0 \ \forall x \in (0, \beta)$ **

- $y > 0$
- for all.

$|\alpha-\beta| = \frac{\sqrt{D}}{|a|}$

$|\alpha-\beta|^2 = \frac{D}{|a|^2} = \frac{D}{a^2}$

$|\alpha-\beta| > 2$

$\# |\alpha-\beta|^2 > 4$

Acidic Question

The graph of $y = ax^2 + bx + c$ is shown in the figure, then which of the following is(are) correct?

D $ab(4a - 2b + c) > 0$

E $b^2 > 4(a^2 + ac)$ $\rightarrow$ $b^2 > 4a^2 + 4ac$

$b^2 - 4ac > 4a^2$

$\# \dfrac{b^2 - 4ac}{a^2} > 4$

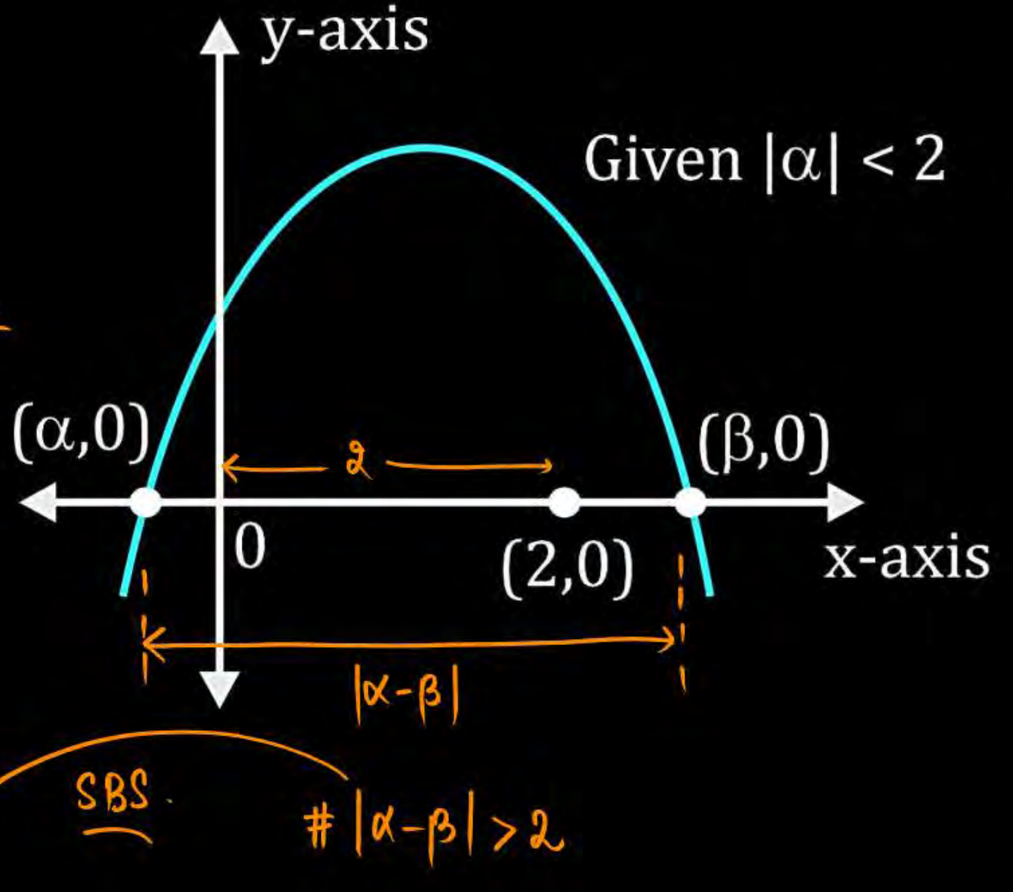
$\dfrac{D}{a^2} > 4$

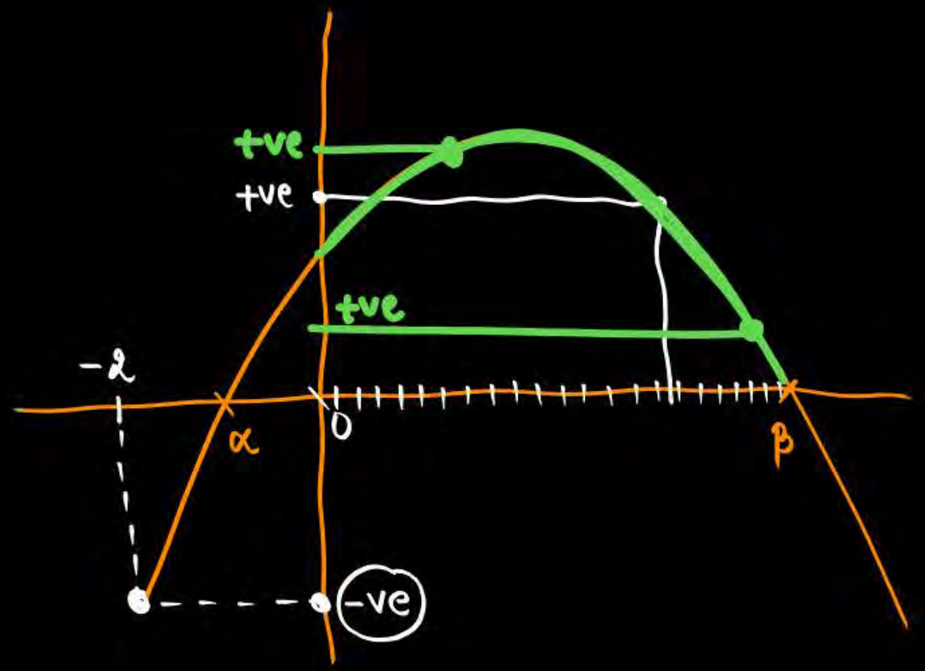
$\# |\alpha - \beta|^2 > 4$

SBS

$\# |\alpha - \beta| > 2$

F $ax^2 + bx + c > 0 \ \forall \ x \in (0, \beta)$





$|\alpha| < 2$

HOW TO READ GRAPH

Answer the following True/False:

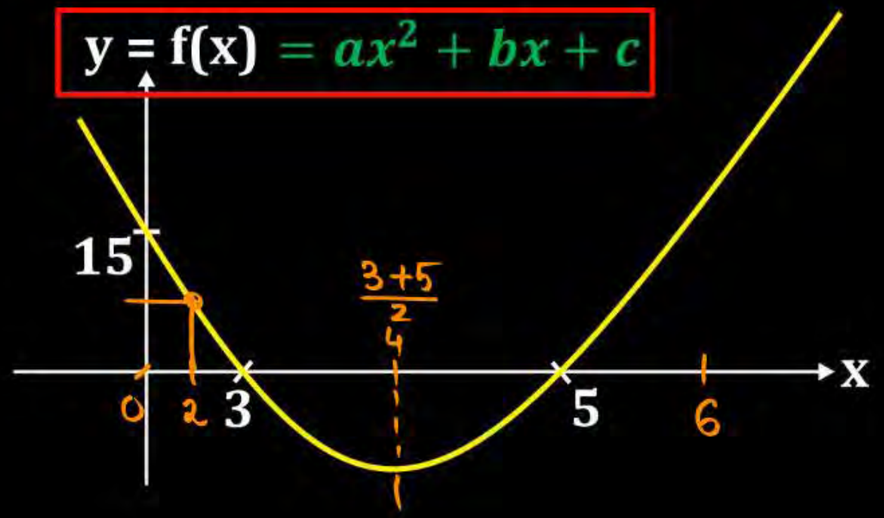
(i) Graph is symmetric about $x = 4$. (T)

(ii) Roots of Quadratic are $x = 3, 5$. (T)

(iii) $f(6) > 0$. → (T)

(iv) $f(2) < 0$. → (F)

(v) $f(0) = 15$ → (T)



HOW TO READ GRAPH

Answer the following True/False:

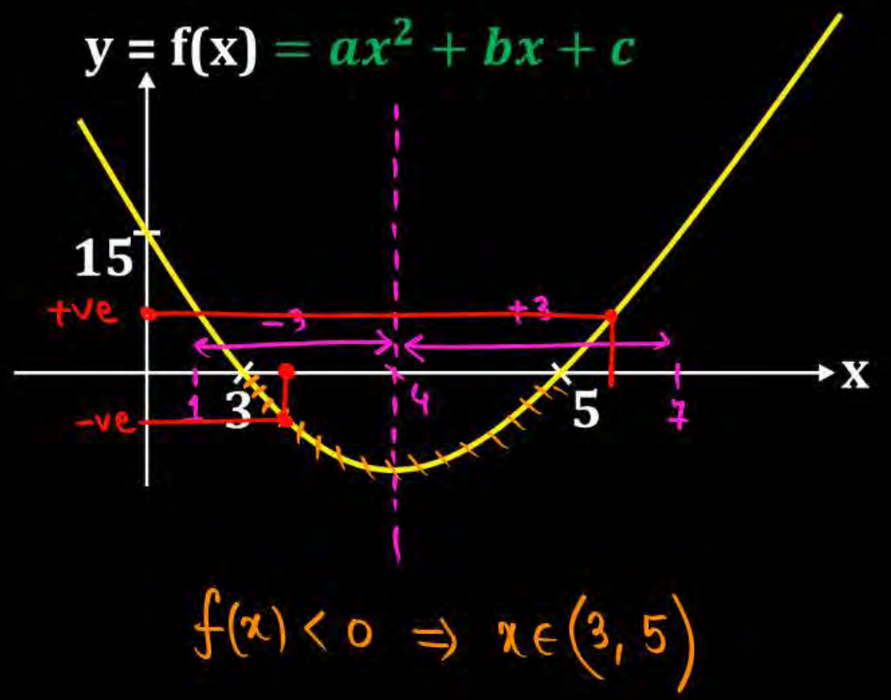
(vi) $f(x) > 0 \; \forall \; x \in (-\infty, 3) \cup (5, \infty)$ (T)

(vii) $f(x) < 0 \; \forall \; x \in [3, 5]$ (F)

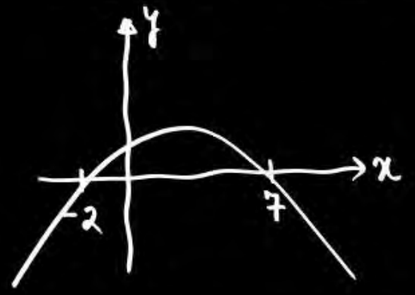
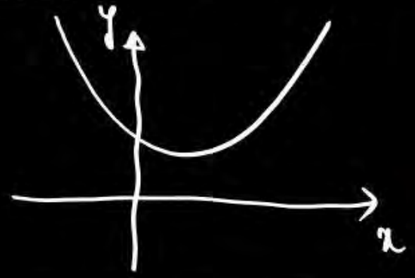
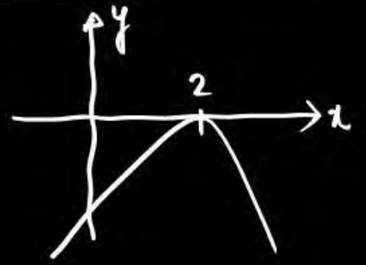
(viii) $f(1) = f(7)$ (T)

(ix) $f(3.25) . f(5.28) > 0$ (F)

-ve +ve

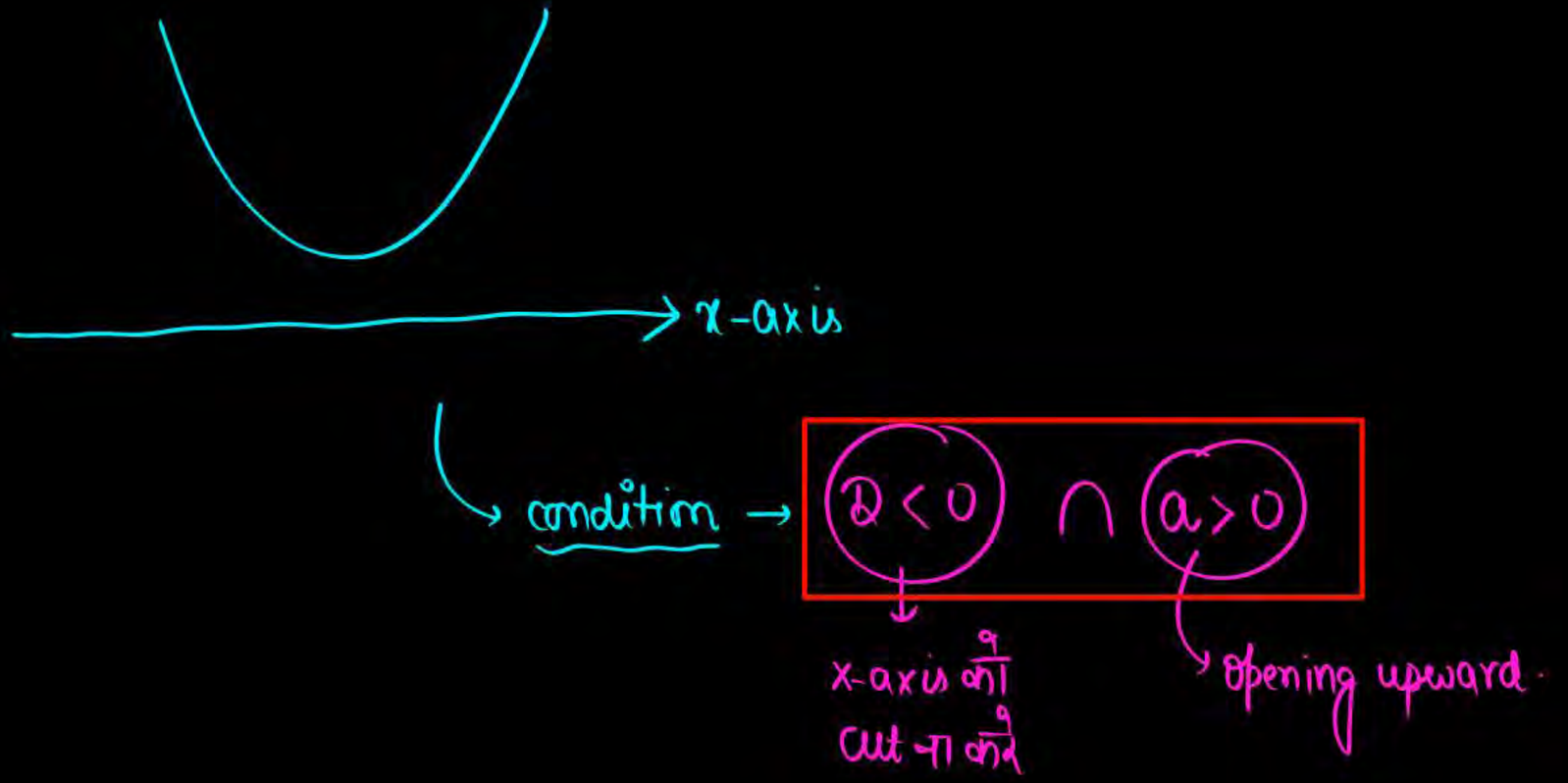


$f(x) < 0 \Rightarrow x \in (3, 5)$

Cases	$f(x) > 0$	$f(x) = 0$	$f(x) \leq 0$
	$x \in (-2, 7)$	$x = -2, 7$	$x \in (-\infty, -2] \cup [7, \infty)$
	हमेशा # $x \in R$	ϕ	ϕ
	ϕ	$x = \alpha$	$x \in R$

THINK!!

What should be the condition such that graph of Quadratic remains always above x-axis??



MOST IMPORTANT NOTE



"Condition"

$ax^2 + bx + c > 0 \; \forall \; x \in R$ ⇒ हमेशा value +ve आएगी ⇒ Graph lies completely above x-axis ⇒



$a > 0 \cap D < 0$

$ax^2 + bx + c < 0 \; \forall \; x \in R$ ⇒ -ve ⇒ below ⇒



$a < 0 \cap D < 0$

$ax^2 + bx + c \geq 0 \; \forall \; x \in R$ ⇒ non-negative आएगी ⇒ Graph either touches x-axis or completely above x-axis

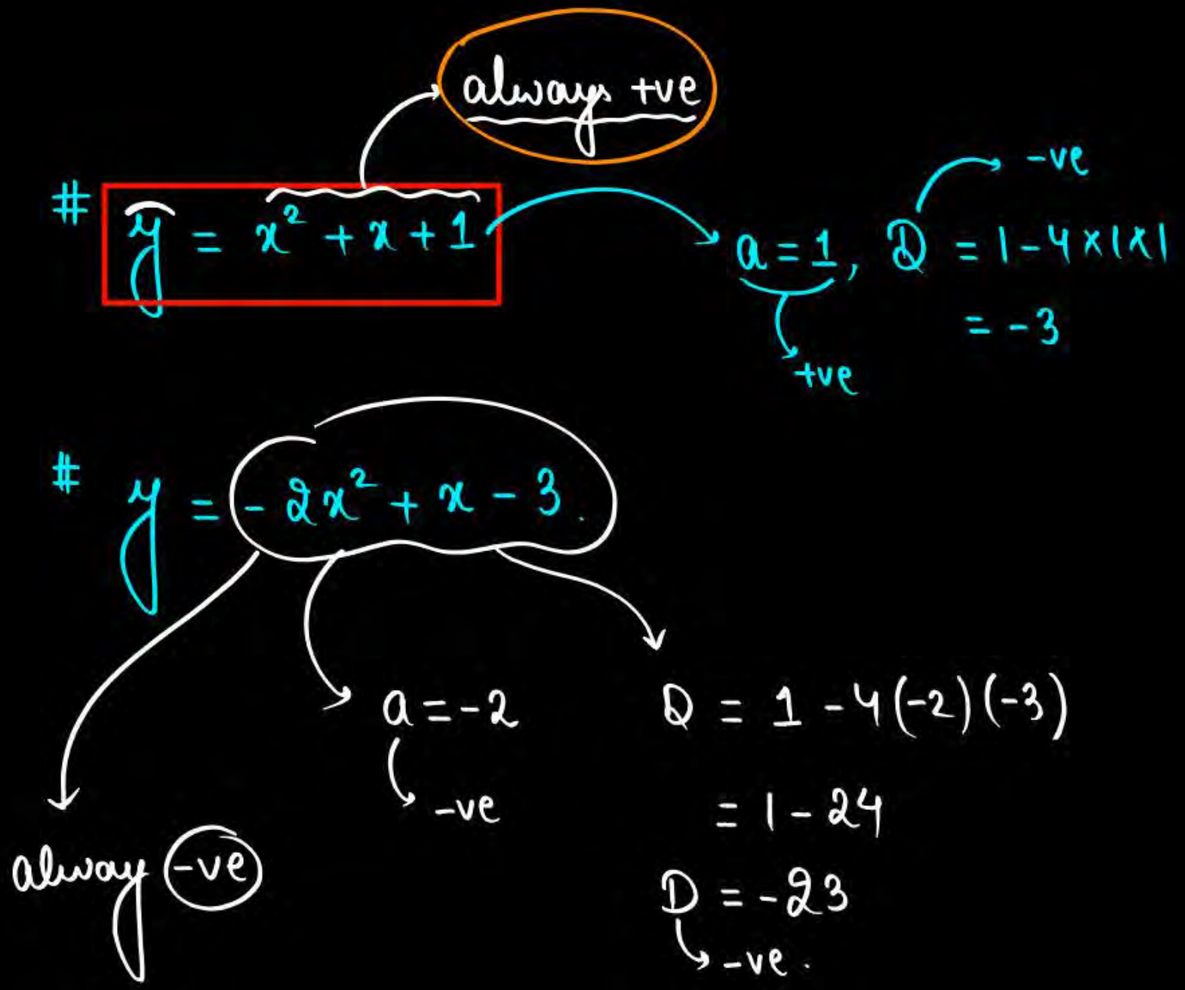


$a > 0 \cap D \leq 0$

$ax^2 + bx + c \leq 0 \; \forall \; x \in R$ ⇒ non-positive ⇒ below ⇒



$a < 0 \cap D \leq 0$



ex:- Solve $\dfrac{(x-3)(x-4)}{x^2+x+1} \geq 0$

x^2+x+1 → No Real Root.

$\Rightarrow (x-3)(x-4) \geq 0$



Question

Find the set of values of a for which the quadratic polynomial

(i) $(a+4)x^2 - 2ax + 2a - 6 \leq 0 \ \forall \ x \in R$

(ii) $(a-1)x^2 - (a+1)x + (a+1) > 0 \ \forall \ x \in R$

हमेशा

$a>0 \cap D<0 \rightarrow$ DIBY – ✗ (Chut!)

(i)



$\Rightarrow a<0 \cap D \leq 0$

$a+4<0$

$a<-4$

$D \leq 0$

$(-2a)^2 - 4(a+4)(2a-6) \leq 0$

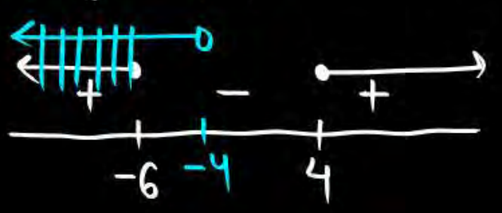
$4a^2 - 4(2a^2+8a-6a-24) \leq 0$

$a^2 - 2a^2 - 2a + 24 \leq 0$

$-a^2 - 2a + 24 \leq 0$

$a^2 + 2a - 24 \geq 0$

$(a+6)(a-4) \geq 0$



$a \in (-\infty, -6]$ Ans.

Acidic Question

Find the set of real values of 'a' for which the graph of the function $f(x) = \dfrac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$ is always below the x-axis.

$$\# \quad \frac{ax^2 + 2(a+1)x + 9a + 4}{1x^2 - 8x + 32} < 0 \quad \forall\ x \in R \quad \Rightarrow \quad ax^2 + 2(a+1)x + 9a + 4 < 0 \quad \forall\ x \in R$$

(−ve)

हमेशा

★ $D < 0$

★ $a < 0 \cap (2(a+1))^2 - 4a(9a+4) < 0$

$4(a^2+2a+1) - 4(9a^2+4a) < 0$

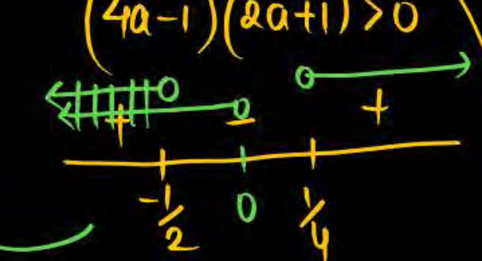
$-8a^2 - 2a + 1 < 0$

$8a^2 + 2a - 1 > 0$

$8a^2 + 4a - 2a - 1 > 0$

$4a(2a+1) - 1(2a+1) > 0$

$(4a-1)(2a+1) > 0$



$a \in (-\infty, -\frac{1}{2})$

for

$\# \quad a > 0, D < 0$

↓

always +ve

$a = 1, \quad D = (-8)^2 - 4 \times 1 \times 32$

$= 64 - 128$

$D = -64$

#S4

Quad. = $y = f(x)$

"Solution"

Quad. > 0 $\forall\ x \in R$ $\Rightarrow$ $a > 0 \cap D < 0$

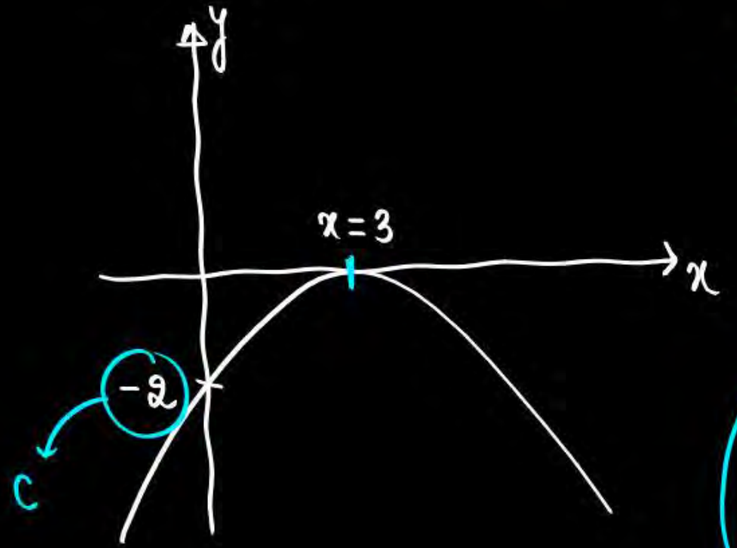
Quad. ≥ 0 $\forall\ x \in R$ $\Rightarrow$ $a > 0 \cap D \leq 0$

Quad. < 0 $\forall\ x \in R$ $\Rightarrow$ $a < 0 \cap D < 0$

Quad. ≤ 0 $\forall\ x \in R$ $\Rightarrow$ $a < 0 \cap D \leq 0$.

Question

If quadratic function $f(x)$ touches x-axis at $x = 3$ and crosses y-axis at $(0, -2)$, then find $f(7)$.



$f(x) = ax^2 + bx + c$.

$\Rightarrow c = -2$ ($c = -2$)

$\Rightarrow D = 0$

$\Rightarrow -\frac{b}{2a} = 3$

$-b = 6a$ or $b = -6a$

$b^2 - 4ac = 0$

$36a^2 - 4a(-2) = 0$

$36a^2 + 8a = 0$

$4a(9a + 2) = 0$

$a = 0, -\frac{2}{9}$ (0 ✗)

$b = -6 \times -\frac{2}{9} = \frac{4}{3}$

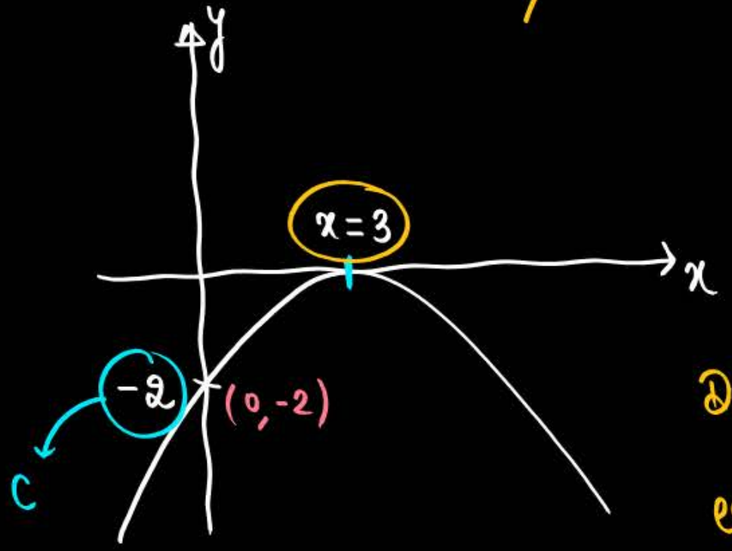
$f(x) = -\frac{2}{9}x^2 + \frac{4}{3}x - 2$.

$x = 7$

$f(7) = -\frac{2}{9} \times 49 + \frac{4}{3} \times 7 - 2$

$= \frac{-98 + 84 - 18}{9}$

$= \frac{-14 - 18}{9} = \left(\frac{-32}{9}\right)$ fns



Sumit:-

$D = 0$

↓

equal root

↳ $x = 3, 3$.

$f(x) = (x-3)(x-3)$

OR

$f(x) = x^2 - (6)x + 9$

$f(7) = 7^2 - 6\times 7 + 9$

$= 7 + 9$

$= 16$ ✗

Santosh:-

$y = k(x^2 - 6x + 9)$

$x = 0, y = -2$

$-2 = k(9)$

$k = -\frac{2}{9}$

$y = -\frac{2}{9}(x^2 - 6x + 9)$

$y = -\frac{2}{9}x^2 + \frac{4}{3}x - 2$ ✓✓

Question [JEE Mains-2020]

Let $f(x)$ be a quadratic polynomial such that $f(-1) + f(2) = 0$. If one of the roots of $f(x) = 0$ is 3, then its other root lies in:



A $(-1, 0)$

B $(-3, -1)$

C $(0, 1)$

D $(1, 3)$

DO IT BY YOURSELF (DIBY)

GRAPH OF QUADRATIC

DIBY-40

Consider the graph $f(x) = ax^2 + bx + c$ and comment on sign of following parts:

(i) (abc) is _____.

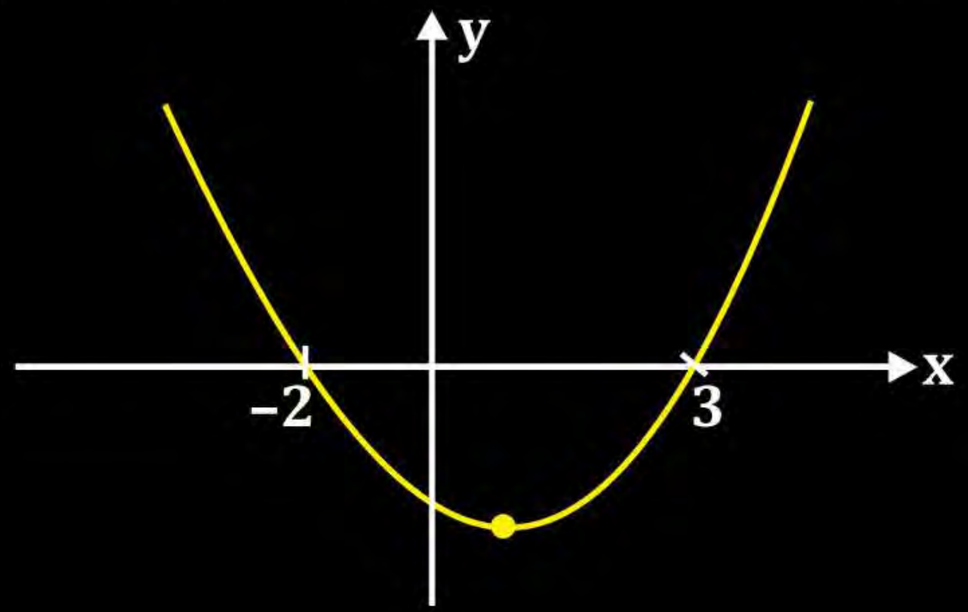
(ii) $(4a + 2b + c)$ is _____.

(iii) for all $x > 3, f(x)$ is _____.

(iv) For all $-2 < x < 3, f(x)$ is _____.

(v) D is _____.

(vi) $(a - b + c)\cdot(9a - 3b + c)$ is _____.



Ans. :

(i) +ve, (ii) −ve, (iii) +ve, (iv) −ve, (v) +ve, (vi) −ve

GRAPH OF QUADRATIC

DIBY-41(Challenger)

Draw the graph of quadratic $(y = ax^2 + bc + c)$ in each part with given conditions:

(i) Roots are 3 & 7 and crosses y-axis at (0, −2)

(ii) Imaginary roots with vertex at $\left(-\frac{3}{4}, \frac{1}{2}\right)$ and $a - b + c = 2$.

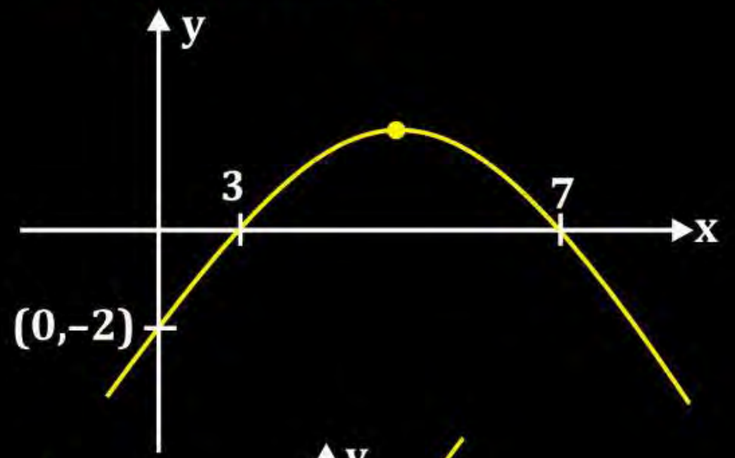
(iii) Touches x-axis at $x = -3$ and value at $x = 2$ is −8.

Also find value of a, b & c in each case.

GRAPH OF QUADRATIC

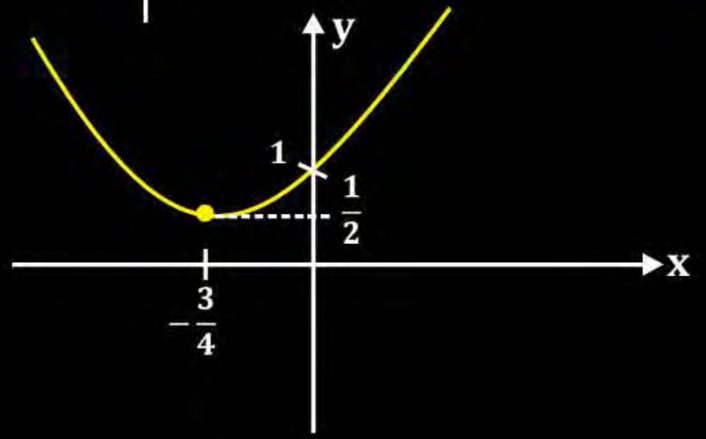
DIBY-41(Answer)

(i)



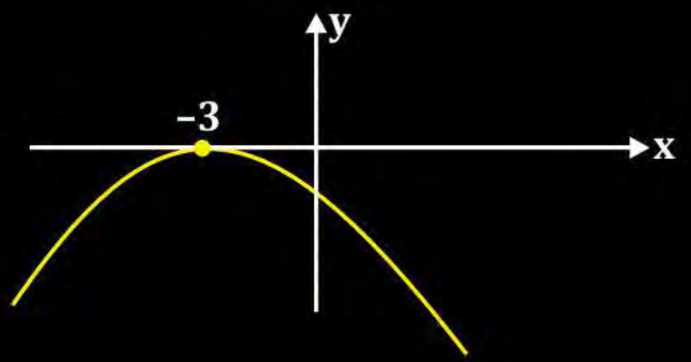
$a = -\frac{2}{21}, b = \frac{20}{21}, c = -2$

(ii)



$a = 24, b = 36, c = 14$

(iii)



$a = \frac{-8}{25}, b = \frac{-48}{25}, c = \frac{-72}{25}$

GRAPH OF QUADRATIC

DIBY-42

If the equation $ax^2 + bc + c = 0, (a > 0)$ has two real roots α and β such that $\alpha < -2$ and $\beta > 2$, then which of the following statements is/are true?

(A) $a - |b| + c < 0$

(B) $c < 0,\ b^2 - 4ac > 0$

(C) $4a - 2|b| + c < 0$

(D) $9a - 3|b| + c < 0$

[Ans. : A, B, C]

DIBY-43

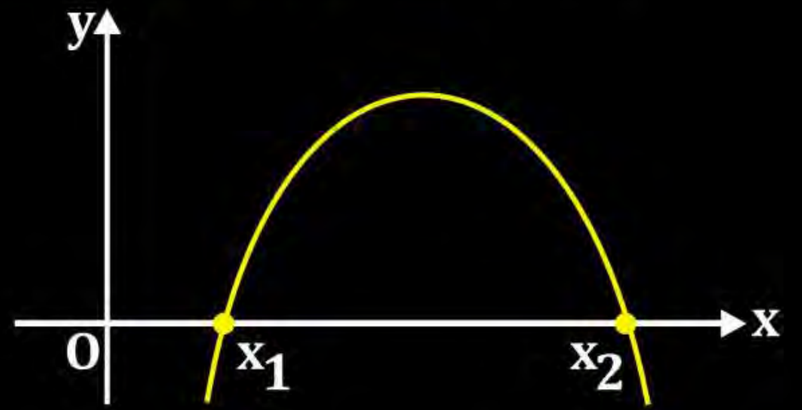
The adjoining figure shows the graph of $y = ax^2 + bx + c$. Then

(A) $a < 0$

(B) $b^2 < 4ax$

(C) $c > 0$

(D) a and b are of opposite sign



[Ans. : A, D]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians